DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING				
Course Code: 24MA313	*Course Title:* ***Engineering Mathematics-III***			
Total Credits: 03 *L-45: T-0: P-0: SL and TW-45*	*SDGs addressed: 4, 5, 9, 16, 17*	*Course Category: Basic Science Course*	*Course Type: Theory*	

Teaching Learning Process		*Total Contact Hours/semester*	*Assessment Weightage and Marks*			
	Lecture	45		CIE	SEE	Total
	Tutorial	-	*Weightage %*	*40*	*60*	*100*
	Practical	-	*Maximum Marks*	*40*	*60*	*100*
	Self-learning and Teamwork	45	*Minimum Marks*	*20*	*25*	*45**
	Total	90				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: Engineering Mathematics-I and Engineering Mathematics-II.

Course Objectives:

Sl. No.	Course Objectives
1	To impart fundamental understanding of probability, statistics, Combinatorics and their applications in various Engineering fields.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Apply least square method to fit a curve for the given data and evaluate the correlation coefficient and regression lines for the data.	L2	WK1, WK3
CO2	Determine the nature of the events and hence calculate the appropriate probabilities of the events.	L2	WK2, WK6
CO3	Classify the random variables to determine the appropriate probability distributions.	L2	WK1, WK4
CO4	Apply the techniques of counting to identify the number of ways in which a given task can be accomplished without list all the possibilities explicitly.	L2	WK7, WK8
CO5	Derive the generating function for the given situation and evaluate the required coefficient.	L2	WK5, WK8

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

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Course Content / Syllabus:

Unit No.	Content	Lecture Hours
1	**Statistics:** Introduction, Definitions, Curve Fitting, equation of Straight line, parabola and exponential, correlation and regression, formula for correlation coefficient, regression lines and angle between the regression lines. **Self-Learning and Teamwork*** • Least square method to derive normal equations. • Fitting of curves: $y = ax^b$, $y = ab^x$. • Find $r,\ \bar{x},\ \bar{y}$ when regression lines are given.	09
2	**Probability Theory:** Basic terminology, Definition of probability, Probability and set notations, Addition law of probability, independent events, conditional probability, multiplication law of probability, Baye's theorem. **Self-Learning and Teamwork*** • Axioms of probability. • Solve logical problems using the theory of probability. • Solve real life problems using Bay's theorem.	09
3	**Random Variables:** Discrete Probability distribution, Continuous Probability distribution, expectation, Variance, Probability generating function, Binomial distribution, Poisson distribution, Normal distribution. **Self-Learning and Teamwork** • Plot Probability density functions for simple distributions. • Visualize how binomial approximates Poisson as n→∞, p→0. • Analyze how small changes in the PDF affect probabilities.	09
4	**Combinatorics-1:** **Fundamental Principles of Counting:** The Rules of Sum and Product, Permutations (linear, circular, identical objects), Combinations – The Binomial Theorem, Multinomial Theorem and Combinations with Repetition. **The Principle of Inclusion and Exclusion**: The Principle of Inclusion and Exclusion, Generalizations of the Principle. **Self-Learning and Teamwork*** • Compare linear vs. circular arrangements with concrete examples. • Compare outputs of $\binom{n}{r}$ and $P(n,r)$ for same values of n and r. • Derive and compute small derangement numbers (e.g., for n = 3, 4, 5), Use PIE to solve problems like - how many numbers from 1 to 100 are not divisible by 2, 3, or 5?	09

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Text Books:

Sl. No.	Author/s	Title	Publisher Details
1	B.S. Grewal	Higher Engineering Mathematics	44th Edition, Khanna Publications, 2015.
2	Ralph P. Grimaldi	Discrete and Combinatorial Mathematics	4th Edition, PHI/Pearson Education, 2005
3	E. Kreyszig	Advanced Engineering Mathematics	10th Edition (Reprint), 2016, John Wiley & Sons.

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	N. P. Bali and Manish Goyal	A Textbook of Engineering Mathematics	Vol. I & II (AICTE), 10th Edition, Laxmi Publications Pvt Ltd, 2019.
2	Alan Tucker	Applied Combinatorics	5th Edition, Wiley-India, 2011
3	Maurice D. Weir, Joel R. Hass and George B. Thomas	Thomas' calculus: Early Transcendentals	12th Edition, Pearson Education, 2016.
4	B. V. Ramana	Higher Engineering Mathematics	11th Edition, Tata-McGraw Hill 2010.
5	Peter V.O 'Neil	Advanced Engineering Mathematics	7th Edition, CENGAGE Learning India Pvt. Ltd. Publishers, 2012.

Web Resources:

Sl. No.	Web Links
1	https://www.classcentral.com/subject/maths
2	https://academicearth.org/

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	2	-	-
CO4	3	1	-	-	-	-	-	-	-	-	-	2	-	-
CO5	3	1	-	-	-	-	-	-	-	-	-	2	-	-

High – 3, Medium – 2, Low – 1

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